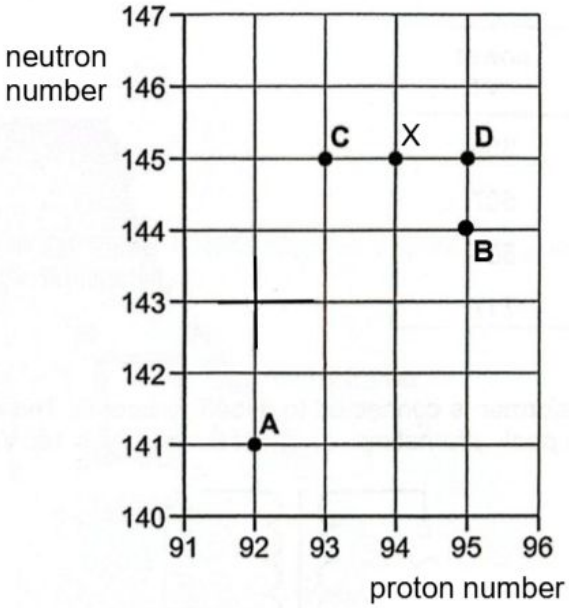
29 The figure shows part of a chart of nuclides where neutron number is plotted against proton number.

An unstable nuclide X decays spontaneously by emitting a β-particle.

Into which nuclide does it decay?



L2 **Answer: B**

β-decay: one neutron disintegrates into one proton and one electron
$proton\ number = 94 + 1 = 95$
$neutron\ number = 145 - 1 = 144$

So the daughter nuclide is:

- **proton number 95 (increases by 1)**
- **neutron number 144 (decreases by 1)**
- **On the graph, this is one square to the right and one square down from X.**

[Turn over